Single-molecule multimodal timing of in vivo mRNA synthesis

JC notes

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1 Abstract

- Poly(A) tail measurement to distinguish transcribing vs. poly(A) transcripts.
- Splicing is much more delayed than previously tought, for at least 10 kb.
- Splicing mostly apparent only in poly(A) transcripts $\rightarrow$ they think it's delayed mostly after 3' end formation.
- Lots of m^6A on unspliced RNA $\rightarrow$ deposition of m^6A happens before splicing.
- m^6A supressed around exon boundaries $\rightarrow$ m^6A topology should be established before EJC deposition.
- 3' end cleavage occurs multiple kbs behind transcription and is coordinated with transcription termination. The 3' end formation is recursive in many genes.
- Overall, they think that cleavage, termination and m^6A deposition occur earlier than most of the splicing.

2 Intro

- Spliceosome assembly is known to be rapid.
- Splicing kinetics: estimates vary across methods.
- Previously, it was tought that EJC prevents m^6A deposition at exon boundaries. Here, they argue that m^6A is placed before splicing (and thus, before EJC placement).
  - m^6A methylation machinery is known to interact with Pol II $\rightarrow$ permitting co-transcriptional methylation.

Poly(A) notes

- Poly(A) site: cannonical AAUAAA motif.
- When Pol II reaches it, it keeps transcribing.
- Protein complex cut RNA there; poly(A) polymerase (PAP) adds ~ 200 As.
- Transcription termination:
  - Torpedo model: Exonuclease XRN2 starts chewing the 5' end of RNA still transcribed by Pol II $\rightarrow$ when it catches Pol II, it knocks it off DNA.
  - Allosteric model: Pol II changes conformation at poly(A) site, becomes less stable and eventually falls of DNA.

3 Results

3.1 Development of a multimodal direct pre-mRNA sequencing technique + dFORCE distinguishes endogenously polyadenylated pre-mRNA

- Need to distinguish whether pre-mRNA has a poly(A) tail:
  - pre-mRNA^{e}: undergoing transcription.
  - pre-mRNA^{a}: polyadenylated pre-mRNA, undergoing post-transcriptional maturation.
- They pulled Pol II with associated pre-mRNA via immunoprecipitation. The pre-mRNA^{a} was still somehow pulled (even thought it was already cleaved - otherwise it would not have the poly(A) tail); probably was somehow attached via the cleavage complex, or some chromatin components.
- They got about 90% of pre-mRNA^{e} and 10% pre-mRNA^{a}.
- 95% of pre-mRNA^{e} have unspliced introns, while this is only 58% for pre-mRNA^{a}.
- Nanopore sequencing requires poly(A) tail $\rightarrow$ added artificially during library preparation. This *in vitro* poly(A) tail is shorter (20-80nt), so they get bimodal distribution of tail lengths, and classify by cut-off.
  - Also, the reads classified as pre-mRNA^{a} have the start of poly(A) highly enriched by known poly(A) sites.

3.2 The landscape of pre-mRNA 3' polyadenylation

- They detect 42k poly(A) sites (PAS) across 11.6k protein-coding genes.
- Detected PAS mostly agreed with annotated PAS, the AAUAAA signal, and were also detected by total RNA-seq.
- About 18% of PAS were in introns (mostly downstream from canonical sites).
- Recursive polyadenylation: reads classified as pre-mRNA^{a}, spanning between two PAS (avg. distance 531 nt). Suggesting that pre-mRNA with poly(A) tail can be cleaved again (at earlier PAS) and also polyadenylated again there.

3.3 Single-molecule kinetics of cleavage and termination

- Use pre-mRNA^{e} extending beyond distal PAS. Classified as cleaved vs. uncleaved, based on their 5' end.
- They model:
  - C_{50}: distance beyond PAS where 50% transcripts are cleaved.
  - T_{50}: distance beyond PAS where density of Pol II active site drop to 50% of coverage downstream of PAS.
- They estimate C_{50} to 2.1kb (lower bound, because of issues with max. read lenght).
- Estimate of T_{50} of 2.3kb.
- When both estimates were available, the median difference was about 95 nt.

Splicing notes

- **Spliceosome**: consists of snRNA and proteins. Recognizes 5' splicing site (GU), 3' site (AG), and **branch point A** (about 20-50nt from the 3' site).
- **Step 1**: Adenosine at branching point attack the 5' site, cutting the exon and forming a covalent bond with the intron → forming a loop (**lariat**).
- **Step 2**: The 5' exon free end attack the 3' site, forming a covalent exon-exon bond, releasing the lariat out.

3.4 Discrepant evidence for co-transcriptional splicing catalysis

- Discrepancy in splicing: pre-mRNA^e ending in exons often show upstream introns as spliced, but pre-mRNA^e ending in introns does not.

3.5 Catalytic intermediates inflate co-transcriptional splicing estimates

- Catalytic intermediates (CI*): ending exactly at 5' exon - after step 1 of splicing, but before step 2 is finished.

3.6 Elongating pre-mRNA is globally unspliced for thousands of nucleotides

- To avoid contamination by CI*, they analyzed pre-mRNA^e ending in the last intron, and also after PAS (but before polyadenylation) $\rightarrow$ about $\sim$ 90% of upstream introns unspliced. Median context of such analysis was 2-3 kb.

3.7 Splicing catalysis is deferred until after polyadenylation

- Pre-mRNA^a was somehow spliced (27% fully, 33% partially, 40% not).

3.8 The quantitative landscape of pre-mRNA m6A

- They basecalled m^{6}A, about 200k sites, mostly known (DR$\underline{A}$CH motif). Using both total RNA (long reads, but without immunoprecipitation) and pre-mRNA.
- On pre-mRNA, m^{6}A was $\sim 17\times$ depleted on introns. Somewhat smooth transition around exon-intron boundaries.
- m^{6}A sites found in pre-mRNA almost always found also in total RNA.